

Strong magnetic fields and contact interactions in few-fermion systems

M. Elyahu, N. Barnea, and J. Kirscher

Technical manual detailing the implementation of a variational solution of the non-relativistic few-body problem in an external, *i.e.*, static magnetic field.

a. *The symmetric Gauge*

$$\mathbf{A}_i = \frac{B_0}{2}(-y(i), x(i), 0) \quad (1)$$

b. *The Hamiltonian*

$$\hat{H} = -\frac{\hbar^2}{2m} \sum_{i=1}^N \left\{ \nabla(i)^2 + i \left(\frac{\hbar^2}{2m} \right) \left(\frac{q(i)B_0}{\hbar} \right) L_z(i) + \left(\frac{\hbar^2}{2m} \right) \left(\frac{q(i)B_0}{\hbar} \right)^2 \frac{1}{4} (x^2(i) + y^2(i)) - g(i) \left(\frac{\hbar^2}{2m} \right) \left(\frac{q(i)B_0}{\hbar} \right) \sigma_z(i) \right\} \quad (2)$$

$$+ \sum_{i < j}^N (C_a + C_b [\sigma^+(i)\sigma^-(j) + \sigma^-(i)\sigma^+(j) - \sigma^z(i)\sigma^z(j)]) e^{-\frac{\Lambda^2}{4}[\mathbf{r}(i)-\mathbf{r}(j)]^2} + \sum_{\text{cyc.}} \sum_{i < j < k} D \cdot e^{-\frac{\Lambda^2}{4}[\mathbf{r}(i)-\mathbf{r}(j)]^2} e^{-\frac{\Lambda^2}{4}[\mathbf{r}(i)-\mathbf{r}(k)]^2} \quad (3)$$

c. *The variational basis*

$$|A, \alpha\rangle := e^{-\frac{1}{2}\mathbf{x}^T A_x \mathbf{x}} e^{-\frac{1}{2}\mathbf{y}^T A_y \mathbf{y}} e^{-\frac{1}{2}\mathbf{z}^T A_z \mathbf{z}} \cdot \sum_{n=1}^{\text{ncmp}} C_\alpha^n |s_1^n, \dots, s_N^n; t_1^n, \dots, t_N^n\rangle \quad (4)$$

with

$$\alpha = 1, \dots, \text{nts_states}$$

d. *The generic matrix element*

$$I_{\mathcal{O}}(A', \alpha', A, \alpha; P) := \left\langle A', \alpha' \mid \hat{\mathcal{O}} \otimes \hat{g} \mid \hat{P}(A), \hat{P}(\alpha) \right\rangle = \left\langle A' \mid \hat{\mathcal{O}} \mid \hat{P}(A) \right\rangle \cdot \left\langle \alpha' \mid \hat{g} \mid \hat{P}(\alpha) \right\rangle \quad (5)$$

with $\hat{P} \in \mathcal{A}$, hence,

$$\hat{P}(A) = T_P^\intercal A T_P := A^P \quad . \quad (6)$$

$$\hat{\mathcal{O}} \in \left\{ \mathbb{1} ; -\sum_{i=1}^N \prod_{c=x,y,z} \boldsymbol{\partial}_c^\intercal(i) \mathbb{1} \boldsymbol{\partial}_c(i) ; \sum_{i=1}^N q(i) L^z(i) ; \sum_{i=1}^N q(i) (\omega_x x^2(i) + \omega_y y^2(i) + \omega_z z^2(i)) ; \sum_{i=1}^N q(i) \sigma^z(i) ; \sum_{i < j}^N e^{-\frac{\Lambda^2}{4}[\mathbf{r}(i)-\mathbf{r}(j)]^2} \right\} \quad (7)$$

e. The matrix elements

$\hat{\mathcal{O}}$	$\left\langle A' \mid \hat{\mathcal{O}} \mid \hat{P}(A) \right\rangle$	$\left\langle \alpha' \mid \hat{\mathcal{G}} \mid \hat{P}(\alpha) \right\rangle$
$\mathbb{1} := \mathbb{1}_{\mathbf{r}}^P \cdot \mathbb{1}_{\mathbf{s}}^P$	$\left(\frac{(2\pi)^{3N}}{\det \mathbb{A}_x \det \mathbb{A}_y \det \mathbb{A}_z} \right)^{\frac{1}{2}}$	$\sum_{n,n'}^{\text{ncmp}(\alpha), \text{ncmp}(\alpha')} C_{\alpha}^n C_{\alpha'}^{n'} \left\langle \mathbf{s}^{n'}; \mathbf{t}^{n'} \mid \hat{P}(\mathbf{s}^n); \hat{P}(\mathbf{t}^n) \right\rangle$
$-\sum_{i=1}^N \prod_{c=x,y,z} \partial_c^{\top}(i) \mathbb{1}_{\partial_c(i)} \cdot \mathbb{1}_{\mathbf{s}}^P$	$\mathbb{1}_{\mathbf{r}}^P \cdot \prod_{c=x,y,z} (A_c)_{im} (\mathbb{A}_c^{-1})_{mn} (A_c^P)_{ni}$	$\mathbb{1}_{\mathbf{s}}^P$
$\sum_{i=1}^N q(i) L_z(i) = q(i) [x(i) \partial_y(i) - y(i) \partial_x(i)]$	0	$\mathbb{Q}_{\mathbf{s}}^P := \sum_{i=1}^N \sum_{n,n'}^{\text{ncmp}(\alpha), \text{ncmp}(\alpha')} [2 - \mathbf{t}^n(\hat{P}(i))] C_{\alpha}^n C_{\alpha'}^{n'} \left\langle \mathbf{s}^{n'}; \mathbf{t}^{n'} \mid \hat{P}(\mathbf{s}^n); \hat{P}(\mathbf{t}^n) \right\rangle$
$\sum_{i=1}^N q(i) (\omega_x x^2(i) + \omega_y y^2(i) + \omega_z z^2(i))$	$\mathbb{1}_{\mathbf{r}}^P \cdot \prod_{c=x,y,z} \omega_c \sum_{i=1}^N (\mathbb{A}_c^{-1})_{ii}$	$\mathbb{Q}_{\mathbf{s}}^P$
$\sum_{i=1}^N q(i) \sigma^z(i)$	$\mathbb{1}_{\mathbf{r}}^P$	$\sum_{i=1}^N \sum_{n,n'}^{\text{ncmp}(\alpha), \text{ncmp}(\alpha')} [2 - \mathbf{t}^n(\hat{P}(i))] [3 - 2\mathbf{s}^n(\hat{P}(i))] C_{\alpha}^n C_{\alpha'}^{n'} \left\langle \mathbf{s}^{n'}; \mathbf{t}^{n'} \mid \hat{P}(\mathbf{s}^n); \hat{P}(\mathbf{t}^n) \right\rangle$
$\sum_{i < j}^N e^{-\frac{\Lambda^2}{4} [\mathbf{r}(i) - \mathbf{r}(j)]^2}$		
$\sum_{\text{cyc. } i < j < k} e^{-\frac{\Lambda^2}{4} [\mathbf{r}(i) - \mathbf{r}(j)]^2} e^{-\frac{\Lambda^2}{4} [\mathbf{r}(i) - \mathbf{r}(k)]^2}$		

(8)

with

$$\mathbb{A}_x = A'_x + A_x^P \quad (9)$$

$$\frac{\Lambda^2}{4} = \mathbf{apot} \quad (10)$$

Spin-isospin wave function for the deuteron:

$$\left| \alpha \hat{=} {}^3S_1; m_S = \begin{Bmatrix} 0 \\ \pm 1 \end{Bmatrix} \right\rangle = \left[\begin{Bmatrix} | \mathbf{12} \rangle + | \mathbf{21} \rangle \\ | \mathbf{11/22} \rangle + | \mathbf{11/22} \rangle \end{Bmatrix} \right] \otimes \left[| \mathbf{pn} \rangle - | \mathbf{np} \rangle \right] \quad (11)$$

$$\mathbf{s}^n(i) = \begin{cases} 1 \Rightarrow \sigma^z(i) = \uparrow \\ 2 \Rightarrow \sigma^z(i) = \downarrow \end{cases} \quad \text{and} \quad \mathbf{t}^n(i) = \begin{cases} 1 \Rightarrow \tau^z(i) = \mathbf{p} \\ 2 \Rightarrow \tau^z(i) = \mathbf{n} \end{cases} \quad (12)$$

#	0 1 b_#	0 2 b_#
0	$\langle \xi_1^{n'}, \dots, \xi_N^{n'} \mid \xi_{\mathfrak{p}_1}^n, \dots, \xi_{\mathfrak{p}_N}^n \rangle$	$\langle \xi_1^{n'}, \dots, \xi_N^{n'} \mid \xi_{\mathfrak{p}_1}^n, \dots, \xi_{\mathfrak{p}_N}^n \rangle$
1	---	$\langle \xi_1^{n'}, \dots, \xi_i^{n'}, \dots, \xi_j^{n'}, \dots, \xi_N^{n'} \mid \xi_{\mathfrak{p}_1}^n, \dots, \xi_{\mathfrak{p}_j}^n, \dots, \xi_{\mathfrak{p}_i}^n, \dots, \xi_{\mathfrak{p}_N}^n \rangle$

TABLE I. Two basic operations are performed by the interaction Hamiltonian Eq.(2), namely, the identity, and the exchange.

Operators.cpp

$$\begin{aligned}
\text{iop} = 0 : & \quad \mathbb{1} & & \text{(Wigner)} & (13) \\
\text{iop} = 1 : & \quad -\frac{1}{4} [\mathbb{1} + \boldsymbol{\tau}(i) \cdot \boldsymbol{\tau}(j)] [\mathbb{1} + \boldsymbol{\sigma}(i) \cdot \boldsymbol{\sigma}(j)] & & \text{(Majorana)} & (14) \\
\text{iop} = 2 : & \quad \frac{1}{2} [\mathbb{1} + \boldsymbol{\sigma}(i) \cdot \boldsymbol{\sigma}(j)] & & \text{(Bartlett)} & (15) \\
\text{iop} = 3 : & \quad -\frac{1}{2} [\mathbb{1} + \boldsymbol{\tau}(i) \cdot \boldsymbol{\tau}(j)] & & \text{(Heisenberg)} & (16)
\end{aligned}$$